

NIRMA UNIVERSITY
Institute of Technology
B.Tech (CSE) and Integrated B.Tech (CSE)
Practical List
Semester-V

3CS501CC24	Design and Analysis of Algorithm	[3-0-2-4]
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Sr. NO	Practical Title	Hours	CLO
1	Topic: Array and Matrix - Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to target. You may assume that each input would have exactly one solution, and you may not use the same element twice. You can return the answer in any order. - You are given an array price where prices[i] is the price of a given stock on the ith day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0. - Given an unsorted array of integers nums, return the length of the longest consecutive elements sequence. You must write an algorithm that runs in O(n) time. - Given an integer array nums, find the subarray with the largest sum, and return its sum. - You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the ith line are (i, 0) and (i, height[i]). Find two lines that together with the x-axis form a container, such that the container contains the most water. Return the maximum amount of water a container can store. Link: https://leetcode.com/problems/two-sum/ https://leetcode.com/problems/best-time-to-buy-and-sell-stock/ https://leetcode.com/problems/longest-consecutive-sequence/ https://leetcode.com/problems/maximum-subarray/ https://leetcode.com/problems/container-with-most-water/	2	3
2	Topic: Sorting Algorithm - Implement iterative and full recursive version of following sorting techniques and repeat the experiment for larger values of n with different input patterns and plot the comparison graph of number of elements versus execution time taken. The elements can be read from a file or can be generated using random number generator. - Selection sort - Insertion sort - Bubble sort	4	2

	2. Given an array of integers nums, sort the array in ascending order and return it. You must solve the problem without using any built-in functions in $O(n \log(n))$ time complexity and with the smallest space complexity possible. Link: https://leetcode.com/problems/sort-an-array/description/		
3	Topic: Searching 1. You are given an $m \times n$ binary matrix mat of 1's (representing soldiers) and 0's (representing civilians). The soldiers are positioned in front of the civilians. That is, all the 1's will appear to the left of all the 0's in each row. 2. you are given a sorted array consisting of only integers where every element appears exactly twice, except for one element which appears exactly once. Return the single element that appears only once. 3. A peak element is an element that is strictly greater than its neighbors. Given a 0-indexed integer array nums, find a peak element, and return its index. If the array contains multiple peaks, return the index to any of the peaks. 4. Given an integer array nums, return an integer array counts where counts[i] is the number of smaller elements to the right of nums[i]. 5. Implement basic operations of Red Black Tree. Links: 1. https://leetcode.com/problems/the-k-weakest-rows-in-a-matrix/description/ 2. https://leetcode.com/problems/single-element-in-a-sorted-array/ 3. https://leetcode.com/problems/find-peak-element/ 4. https://leetcode.com/problems/kth-smallest-number-in-multiplication-table/description/ 5. https://leetcode.com/problems/reverse-nodes-in-k-group/ 6. https://leetcode.com/problems/count-of-smaller-numbers-after-self/	2	3
4	Topic: Recursion 1. Given an integer n, return true if it is a power of four. Otherwise, return false. An integer n is a power of four, if there exists an integer x such that $n == 4^x$ 2. The Fibonacci numbers, commonly denoted $F(n)$ form a sequence, called the Fibonacci sequence, such that each number is the sum of the two preceding ones, starting from 0 and 1. 3. Given an array of distinct integers candidates and a target integer target, return a list of all unique combinations of candidates where the chosen numbers sum to target. You may return the combinations in any order. 4. Given a collection of candidate numbers (candidates) and a target number (target), find all unique combinations in candidates where the candidate numbers sum to target. 5. Given a string expression of numbers and operators, return all possible results from computing all the different possible ways to group numbers and operators. You may return the answer in any order. Links: 1. https://leetcode.com/problems/power-of-four/ 2. https://leetcode.com/problems/fibonacci-number/ 3. https://leetcode.com/problems/combination-sum/ 4. https://leetcode.com/problems/combination-sum-ii/ 5. https://leetcode.com/problems/different-ways-to-add-parentheses/	2	3

5	Topic: Linked List and Binomial Heap - Given the head of a singly linked list, return the middle node of the linked list. If there are two middle nodes, return the second middle node. - Given an integer array <code>nums</code> sorted in non-decreasing order, remove the duplicates in-place such that each unique element appears only once. The relative order of the elements should be kept the same. Then return the number of unique elements in <code>nums</code>. - Implement binomial heap and all operations (Not on LeetCode). Links: https://leetcode.com/problems/middle-of-the-linked-list/ https://leetcode.com/problems/remove-duplicates-from-sorted-array	2	4
6	Topic: Divide and conquer - Given an array <code>nums</code> of size <code>n</code>, return the majority element. The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array. - Reverse bits of a given 32 bits unsigned integer. - Your task is to calculate $ab \bmod 1337$ where <code>a</code> is a positive integer and <code>b</code> is an extremely large positive integer given in the form of an array. - Given the head of a singly linked list where elements are sorted in ascending order, convert it to a height-balanced binary search tree. - Given an integer array <code>nums</code>, return the number of reverse pairs in the array. Links: https://leetcode.com/problems/majority-element/ https://leetcode.com/problems/reverse-bits/ https://leetcode.com/problems/super-pow/ https://leetcode.com/problems/convert-sorted-list-to-binary-search-tree/ https://leetcode.com/problems/reverse-pairs/	4	3
7	Topic: Greedy algorithms - Given an integer array <code>nums</code> of $2n$ integers, group these integers into <code>n</code> pairs $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)$ such that the sum of $\min(a_i, b_i)$ for all <code>i</code> is maximized. Return the maximized sum. - Given a string <code>s</code>, return true if the <code>s</code> can be palindrome after deleting at most one character from it. - There are <code>n</code> points on a road you are driving your taxi on. The <code>n</code> points on the road are labeled from 1 to <code>n</code> in the direction you are going, and you want to drive from point 1 to point <code>n</code> to make money by picking up passengers. You cannot change the direction of the taxi. - You are assigned to put some amount of boxes onto one truck. You are given a 2D array <code>boxTypes</code>, where <code>boxTypes[i] = [numberOfBoxes_i, numberOfUnitsPerBox_i]</code>. - You are given an integer <code>n</code>. There are <code>n</code> rooms numbered from 0 to <code>n - 1</code>. Links: https://leetcode.com/problems/array-partition/ https://leetcode.com/problems/valid-palindrome-ii/ https://leetcode.com/problems/maximum-number-of-events-that-can-be-attended/	4	4

	4. https://leetcode.com/problems/maximum-units-on-a-truck/ 5. https://leetcode.com/problems/meeting-rooms-iii/		
8	Topic: Tree and Branch & Bound Algorithm 1. There is a binary tree rooted at 0 consisting of n nodes. The nodes are labelled from 0 to n - 1. You are given a 0-indexed integer array parents representing the tree, where parents[i] is the parent of node i. Since node 0 is the root, parents[0] == -1. 2. There is an undirected tree with n nodes labeled from 0 to n - 1 and n - 1 edges. 3. A path in a binary tree is a sequence of nodes where each pair of adjacent nodes in the sequence has an edge connecting them. A node can only appear in the sequence at most once. Note that the path does not need to pass through the root 4. Given a list of words, list of single letters (might be repeating) and score of every character. Return the maximum score of any valid set of words formed by using the given letters (words[i] cannot be used two or more times).. Links: 1. https://leetcode.com/problems/count-nodes-with-the-highest-score/ 2. https://leetcode.com/problems/reachable-nodes-with-restrictions/ 3. https://leetcode.com/problems/binary-tree-maximum-path-sum/ 4. https://leetcode.com/problems/maximum-score-words-formed-by-letters/description/ (Branch and Bound)	2	3
9	Topic: Dynamic Programming 1. You are given an array price where prices[i] is the price of a given stock on the ith day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. 2. Given an integer array nums, return the length of the longest strictly increasing subsequence. 3. Given two strings word1 and word2, return the minimum number of operations required to convert word1 to word2. 4. Given a sequence of matrices, find the most efficient way to multiply these matrices together. The efficient way is the one that involves the least number of multiplications. The dimensions of the matrices are given in an array arr[] of size N (such that N = number of matrices + 1), where the i th matrix has the dimensions (arr[i-1] x arr[i]). 5. You are given an integer array coins representing coins of different denominations and an integer amount representing a total amount of money. Return the number of combinations that make up that amount. If that amount of money cannot be made up by any combination of the coins, return 0. Links: 1. https://leetcode.com/problems/best-time-to-buy-and-sell-stock/ 2. https://leetcode.com/problems/longest-increasing-subsequence/ 3. https://leetcode.com/problems/edit-distance/ 4. https://www.geeksforgeeks.org/dsa/matrix-chain-multiplication-dp-8/ 5. https://leetcode.com/problems/coin-change-ii/description/	4	4
10	Topic: Graph and Backtracking Algorithm 1. In a town, there are n people labeled from 1 to n. There is a rumor that one of these people is secretly the town judge.	4	4

	2. There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisite where prerequisites[i] = [ai, bi] indicates that you must take course bi first if you want to take course ai. 3. There is an undirected graph with n nodes, where each node is numbered between 0 and n - 1. You are given a 2D array graph, where graph[u] is an array of nodes that node u is adjacent to. More formally, for each v in graph[u], there is an undirected edge between node u and node v. 4. There are n cities. Some of them are connected, while some are not. If city a is connected directly with city b, and city b is connected directly with city c, then city a is connected indirectly with city c. 5. The n-queens puzzle is the problem of placing n queens on an n x n chessboard such that no two queens attack each other. Given an integer n, return all distinct solutions to the n-queens puzzle. You may return the answer in any order. Links: 1. https://leetcode.com/problems/find-the-town-judge/ 2. https://leetcode.com/problems/course-schedule/ 3. https://leetcode.com/problems/is-graph-bipartite/ 4. https://leetcode.com/problems/number-of-provinces/ 5. https://leetcode.com/problems/n-queens/description/		
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